

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\mathcal{I}_D(\delta_1, \delta_2) - \mathcal{I}_D(0, 0) = \sup_{\tilde{\gamma} \in \Sigma_D(\delta)} \int_{\mathcal{V}} g \, d\tilde{\gamma} - \mathcal{I}_D(0, 0) \leq \Psi(\delta_1 + \epsilon, \delta_2 + \epsilon).$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$S_f = -\frac{1}{2g^2} Tr \left(\sum_{i=0}^3 \overline{\psi_i} \Gamma^\mu [A_\mu, \psi_i] + 2 \sum_{i=1}^3 \overline{(\psi_i)^c} \bar{\Gamma}^{(i)} [B_i^\dagger, \psi_0] + \sum_{i,j,k=1}^3 |\epsilon_{ijk}| \overline{(\psi_i)^c} \Gamma^{(j)} [B_j, \psi_k] + h.c. \right)$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\det_{(i,j)} \left(\frac{1}{\tilde{w}_i - \tilde{z}_j} \right) = (-1)^{\frac{n(n-1)}{2}} \frac{\prod_{1 \leq i < j \leq n} (\tilde{w}_i - \tilde{w}_j)(\tilde{z}_i - \tilde{z}_j)}{\prod_{i,j=1}^n (\tilde{w}_i - \tilde{z}_j)}.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$U_{\text{AP}}^{\text{bf}}(\vec{x}; C) = e^{-i\frac{\theta}{2}\tau^3} U_{\text{AP}}^{\text{bf}}(D(e^{i\frac{\theta}{2}\tau^3}) \cdot \vec{x}; e^{i\frac{\theta}{2}\tau^3} C e^{-i\frac{\theta}{2}\tau^3}) e^{i\frac{\theta}{2}\tau^3}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$[\mathbf{d}_\alpha, \mathbf{d}_\beta] = \epsilon_{\alpha\beta}{}^\gamma \mathbf{d}_\gamma, \quad [\mathbf{d}_\alpha, \mathbf{s}_m^a] = \mathbf{s}_m^b (\sigma_\alpha)_b{}^a, \quad \{\mathbf{s}_m^a, \mathbf{s}_m^b\} = -m^2 (\sigma^\alpha)^{ab} \mathbf{d}_\alpha,$$